Quadratics & Complex Numbers

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 8$ in factoring polynomials completely, evaluate $f(5)$.

Final Answer: 28. Substitute $x = 5$: $f(5) = 4(5) + 8 = 28$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 9$ in review: graphs and characteristics of quadratic functions, evaluate $f(2)$.

Final Answer: 19. Substitute $x = 2$: $f(2) = 5(2) + 9 = 19$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for review: graphs and characteristics of quadratic functions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 20.5. The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 3$ in review: solving quadratic equations by factoring, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 6(2) + 3 = 15$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for review: solving quadratic equations by factoring consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 4$ in complex numbers, evaluate $f(2)$.

Final Answer: 8. Substitute $x = 2$: $f(2) = 2(2) + 4 = 8$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for complex numbers consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Quadratics & Complex Numbers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Solve $x^2 - 4x + 13 = 0$ and express roots in the form $a \pm bi$.

Discriminant $= 16 - 52 = -36$. $x = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Solve the exponential equation $4^{x+1} - 5(2^x) + 1 = 0$ for all real values of x.

Rewrite the first term in terms of base 2: $4^{x+1} = 4^1 \cdot (2^2)^x = 4 \cdot (2^x)^2$. Substitute $u = 2^x$ (with constraint $u > 0$): $4u^2 - 5u + 1 = 0$. Factor the quadratic: $(4u - 1)(u - 1) = 0 \Rightarrow u = \frac{1}{4}$ or $u = 1$. Back-substitute $2^x = u$:$2^x = \frac{1}{4} = 2^{-2} \Rightarrow x = -22^x = 1 = 2^0 \Rightarrow x = 0$Both solutions are real and valid: $x = -2$ and $x = 0$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

In an arithmetic sequence, the 4th term is 17 and the 10th term is 47. Find the first term a_1, the common difference d, and the sum of the first 25 terms S_{25}.

Using $a_n = a_1 + (n - 1)d$: $a_4 = a_1 + 3d = 17$, $a_{10} = a_1 + 9d = 47$. Subtract the first equation from the second: $6d = 30 \Rightarrow d = 5$. Substitute $d = 5$ into the first equation: $a_1 + 3(5) = 17 \Rightarrow a_1 = 17 - 15 = 2$. Find the sum of the first 25 terms using $S_n = \frac{n}{2}[2a_1 + (n - 1)d]$:

$S_{25} = \frac{25}{2}[2(2) + 24(5)] = \frac{25}{2}[4 + 120] = \frac{25}{2}(124) = 25 \times 62 = 1550$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Write the general circle equation $x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form, identify its center and radius, and find the equation of the tangent line to the circle at the point $P(4, 1)$.

Complete the square for x and y: $(x^2 - 8x + 16) + (y^2 + 6y + 9) = -9 + 16 + 9 = 16$. $(x - 4)^2 + (y + 3)^2 = 16$. Center is $C(4, -3)$ and radius is $r = \sqrt{16} = 4$. Check the point $P(4, 1)$: $(4 - 4)^2 + (1 + 3)^2 = 0 + 16 = 16$ ✓. The radius connects $C(4, -3)$ to $P(4, 1)$, which is a vertical line $x = 4$. Therefore, the tangent line through $P(4, 1)$ is horizontal: $y = 1$.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve $\sqrt{3x+1} = x-1$, checking for extraneous solutions.

Square both sides: $3x+1 = (x-1)^2 = x^2-2x+1 \Rightarrow x^2-5x = 0 \Rightarrow x = 0$ or $x = 5$. Check $x = 0$: $1 = -1$ ✗ (extraneous). Check $x = 5$: $4 = 4$ ✓. Solution: $x = 5$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Write $9x^2 + 4y^2 - 36x + 8y + 4 = 0$ in standard form and identify the conic.

Complete the square: $9(x-2)^2 + 4(y+1)^2 = 36 \Rightarrow \frac{(x-2)^2}{4} + \frac{(y+1)^2}{9} = 1$. Ellipse, centre $(2,\ -1)$, $a = 3$, $b = 2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

Sketch $y = \frac{2x+1}{x-3}$, showing all asymptotes and intercepts.

Vertical asymptote: $x = 3$. Horizontal asymptote: $y = 2$ (leading coefficients). y-intercept: $y = -1/3$. x-intercept: $x = -1/2$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Conic Sections

Solve $2^{x+1} = 5^{x-1}$, giving your answer to 3 significant figures.

Take logs: $(x+1)\ln 2 = (x-1)\ln 5$. $x\ln 2 - x\ln 5 = -\ln 5 - \ln 2$. $x = \frac{-\ln 10}{\ln 2 - \ln 5} = \frac{-\ln 10}{\ln(2/5)} \approx 2.48$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Trigonometric Functions

[Conic Sections · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions & Their Graphs

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Matrices & Systems

[Functions & Their Graphs · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Probability & Statistics

[Matrices & Systems · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 6$ in review: factoring techniques, evaluate $f(2)$.

a. 10 ✓ **b.** 12

c. 4 **d.** 14

Substitute $x = 2$: $f(2) = 2(2) + 6 = 10$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 7$ in factoring with sum and difference of cubes, evaluate $f(2)$.

a. 13 ✓ **b.** 16

c. 6 **d.** 17

Substitute $x = 2$: $f(2) = 3(2) + 7 = 13$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: factoring techniques consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for factoring with sum and difference of cubes consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine the nature of the roots of $x^2 + 3x + 4 = 0$.

a. Two distinct real roots

b. One repeated root

c. Four roots

d. **Two non-real complex roots** ✓

Applying Quadratics & Complex Numbers principles, $\Delta = 9 - 16 = -7 < 0$. Therefore, the verified correct option is Two non-real complex roots.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let m be a real parameter. Consider the quadratic equation $z^2 - 2mz + (m^2 + 3m - 6) = 0$ with complex roots z_1 and z_2. In the complex plane, the origin $O(0)$ and the points representing z_1 and z_2 form the vertices of a non-degenerate equilateral triangle. What is the sum of the areas of all possible equilateral triangles that can be formed in this way?

a. $15\sqrt{3}$ ✓

b. $12\sqrt{3}$

c. $\frac{15\sqrt{3}}{2}$

d. $9\sqrt{3}$

We solve the problem by synthesizing Vieta's formulas with the geometric properties of complex numbers. **Step 1: Condition for an Equilateral Triangle with the Origin** In the complex plane, three points 0, z_1, and z_2 form the vertices of an equilateral triangle if and only if one non-zero vertex is obtained by rotating the other about the origin by $\pm\frac{\pi}{3}$: $\frac{z_2}{z_1} = e^{\pm i\pi/3} = \frac{1 \pm i\sqrt{3}}{2}$ This is equivalent to the quadratic in $\frac{z_2}{z_1}$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the quadratic equation $z^2 - (3 + 2i)z + (1 + 3i) = 0$. Let the roots of this equation be α and β. If the complex number w is defined such that $w = \frac{\alpha^2 + \beta^2}{2}$, find the modulus of w, $|w|$.

a. $\sqrt{5}$ ✓

b. $\frac{\sqrt{5}}{2}$

c. $\sqrt{29}$

d. $\frac{\sqrt{29}}{2}$

The given quadratic equation is $z^2 - (3 + 2i)z + (1 + 3i) = 0$. By Vieta's formulas, the sum of the roots is $\alpha + \beta = 3 + 2i$ and the product of the roots is $\alpha\beta = 1 + 3i$. We need to find $w = \frac{\alpha^2 + \beta^2}{2}$. We can express $\alpha^2 + \beta^2$ in terms of the sum and product of the roots: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$. Substituting the values from Vieta's formulas: $\alpha^2 + \beta^2 = (3 + 2i)^2 - 2(1 + 3i)\ \alpha^2 + \beta^2 = (9 + 12i + 4i^2) - (2 + 6i)$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the quadratic equation $z^2 - (3 + i)z + (2 + 2i) = 0$, where z is a complex number. Let the roots of this equation be z_1 and z_2. Determine the value of $|z_1 - z_2|^2$.

a. 1

b. 2

c. **4** ✓

d. 5

Let the quadratic equation be $az^2 + bz + c = 0$. The roots are z_1 and z_2. We know that $z_1 + z_2 = -\frac{b}{a}$ and $z_1 z_2 = \frac{c}{a}$. The discriminant of the quadratic equation is $\Delta = b^2 - 4ac$. The roots can be expressed as $z_1 = \frac{-b + \sqrt{\Delta}}{2a}$ and $z_2 = \frac{-b - \sqrt{\Delta}}{2a}$. Therefore, $z_1 - z_2 = \frac{\sqrt{\Delta}}{a}$. Consequently, $|z_1 - z_2|^2 = \left|\frac{\sqrt{\Delta}}{a}\right|^2 = \frac{|\Delta|}{|a|^2}$. In our given equation.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Consider the quadratic equation $2x^2 - (k-1)x + (k+1) = 0$ where k is a real constant. The equation has two non-real complex conjugate roots. If the imaginary parts of the roots have a magnitude of $\frac{\sqrt{7}}{2}$, what is the greatest possible value of k?

a. **7** ✓

b. 3

c. 10

d. 5

To find the parameter k, apply the quadratic formula to $ax^2 + bx + c = 0$: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ Identify the coefficients from the given equation: $a = 2, \quad b = -(k-1), \quad c = k+1$ Compute the discriminant Δ: $\Delta = [-(k-1)]^2 - 4(2)(k+1) = (k^2 - 2k + 1) - (8k + 8) = k^2 - 10k - 7$ Since the equation has non-real complex roots, $\Delta 0$, confirming the roots are non-real. The greatest possible value of k is 7. - Misconception 3 arises from selecting the smalle.

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Which of the following is equivalent to the expression $(3 + 4i)(2 - 5i)$, where $i = \sqrt{-1}$?

a. $26 - 7i$ ✓

b. $26 + 7i$

c. $-14 - 7i$

d. $6 - 20i$

To multiply two complex numbers, expand using the distributive property (FOIL): $(3 + 4i)(2 - 5i) = (3)(2) + (3)(-5i) + (4i)(2) + (4i)(-5i)$ Compute each term: $= 6 - 15i + 8i - 20i^2$ Combine the imaginary parts: $-15i + 8i = -7i$ Recall that by definition, $i^2 = -1$. Substitute -1 for i^2: $-20i^2 = -20(-1) = +20$ Combine the real parts: $6 + 20 = 26$ Thus, the expression simplifies to: $26 - 7i$ Common Misconceptions: - $26 + 7i$ results from an arithmetic sign error when combining $-15i + 8i$. - $-14 - 7i$.

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

If $z^2 = \bar{z}$, find all solutions.

a. $z = 0$

b. $z = 0, 1, e^{\pm 2\pi i/3}$ ✓

c. $z = 0, 1$

d. $z = \pm 1$

Write $z = re^{i\theta}$: $r^2 e^{2i\theta} = re^{-i\theta} \Rightarrow r^2 = r \Rightarrow r = 0$ or 1. If $r = 1$: $e^{3i\theta} = 1 \Rightarrow \theta = 0, 2\pi/3, 4\pi/3$. Solutions: $0, 1, e^{2\pi i/3}, e^{4\pi i/3}$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Express $\frac{1+\cos\theta+i\sin\theta}{1+\cos\theta-i\sin\theta}$ in polar form.

a. $e^{i\theta}$ ✓

b. 1

c. $e^{2i\theta}$

d. $\cos\theta+i\sin\theta$ only for $\theta=0$

Numerator: $2\cos^2(\theta/2)+2i\sin(\theta/2)\cos(\theta/2)=2\cos(\theta/2)e^{i\theta/2}$. Denominator: $2\cos(\theta/2)e^{-i\theta/2}$. Ratio: $e^{i\theta}$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the axis of symmetry and vertex of $y=-3x^2+12x-1$.

a. Axis: $x=-2$; vertex $(-2,11)$

b. **Axis: $x=2$; vertex $(2,11)$** ✓

c. Axis: $x=4$; vertex $(4,1)$

d. Axis: $x=2$; vertex $(2,-1)$

Applying Quadratics & Complex Numbers principles, $x=-12/(-6)=2$. $y=-12+24-1=11$. Therefore, the verified correct option is Axis: $x=2$; vertex $(2,11)$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which complex number is its own conjugate?

a. $-i$

b. **Any real number** ✓

c. $1+i$

d. i

Applying Quadratics & Complex Numbers principles, Conjugate of $a+0i$ is $a-0i=a$. Therefore, the verified correct option is Any real number.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the product of roots of $3x^2-7x+2=0$.

a. 7

b. 2

c. $\frac{2}{3}$ ✓

d. $\frac{7}{3}$

Applying Quadratics & Complex Numbers principles, By Vieta: product $=c/a=2/3$. Therefore, the verified correct option is $\frac{2}{3}$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum of roots of $3x^2 - 7x + 2 = 0$.

a. $\frac{2}{3}$

b. 3

c. 7

d. $\frac{7}{3}$ ✓

Applying Quadratics & Complex Numbers principles, By Vieta: sum $= -b/a = 7/3$. Therefore, the verified correct option is $\frac{7}{3}$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A ball thrown upward has height $h(t) = -4t^2 + 16t + 5$. Find the maximum height.

a. 16

b. 5

c. 20

d. 21 ✓

Applying Quadratics & Complex Numbers principles, Vertex: $t = 2$. $h(2) = -16 + 32 + 5 = 21$. Therefore, the verified correct option is 21.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions & Their Graphs

Find the image of the circle $|z| = 2$ under the map $w = z + 1/z$.

a. **An ellipse** ✓

b. A line

c. A parabola

d. A circle of radius 3

$z = 2e^{i\theta} \Rightarrow w = 2e^{i\theta} + \frac{1}{2}e^{-i\theta} = \frac{5}{2}\cos\theta + i\frac{3}{2}\sin\theta$: parametric ellipse.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Matrices & Systems

Find the minimum value of $f(x) = x^2 - 6x + 10$.

a. 1 ✓

b. -1

c. 6

d. 10

Applying Quadratics & Complex Numbers principles, Vertex at $x = 3$: $f(3) = 9 - 18 + 10 = 1$. Therefore, the verified correct option is 1.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Probability & Statistics

A quadratic with roots $r_1 = 2$ and $r_2 = 5$ has equation:

a. $x^2 - 3x + 10 = 0$

b. $x^2 - 10x + 7 = 0$

c. $x^2 - 7x + 10 = 0$ ✓

d. $x^2 + 7x + 10 = 0$

Applying Quadratics & Complex Numbers principles, $(x - 2)(x - 5) = x^2 - 7x + 10$. Therefore, the verified correct option is $x^2 - 7x + 10 = 0$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Polynomial Functions

If $z_1 = 2\angle 45°$ and $z_2 = 3\angle 30°$ (polar form), find $|z_1 z_2|$.

a. $6\angle 75°$

b. 5

c. $\sqrt{13}$

d. 6 ✓

Applying Quadratics & Complex Numbers principles, $|z_1 z_2| = |z_1||z_2| = 2 \cdot 3 = 6$. Therefore, the verified correct option is 6.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Rational Expressions

Convert to standard form: $(3 + 4i)(1 - 2i)$.

a. $3 - 8i$

b. $11 - 2i$ ✓

c. $11 + 2i$

d. $7 + 2i$

Applying Quadratics & Complex Numbers principles, $3 - 6i + 4i - 8i^2 = 3 - 2i + 8 = 11 - 2i$. Therefore, the verified correct option is $11 - 2i$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the two square roots of -9.

a. ± 3

b. $\pm 3i$ ✓

c. $3i$ only

d. $\pm\sqrt{-9} = \pm 3$

Applying Quadratics & Complex Numbers principles, $x^2 = -9 \Rightarrow x = \pm 3i$. Therefore, the verified correct option is $\pm 3i$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Which is NOT a quadratic equation?

a. $2x^2 = 5$

b. $x^2 + x + 1 = 0$

c. $x^2 - 1 = 0$

d. $x^3 + 2x = 0$ ✓

Applying Quadratics & Complex Numbers principles, Cubic, not quadratic. Therefore, the verified correct option is $x^3 + 2x = 0$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

The discriminant of $2x^2 + 3x + 5$ is:

a. -31 ✓

b. -49

c. 31

d. 49

Applying Quadratics & Complex Numbers principles, $9 - 4(2)(5) = 9 - 40 = -31$. Therefore, the verified correct option is -31.

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